The Heat Equation is "Flat"

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under the supervision of

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Introduction

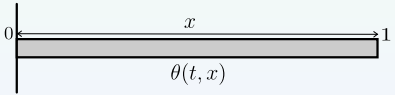
A Minion Thanks

To The Organizers



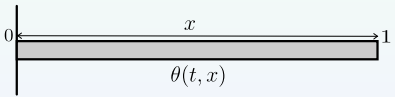
The Heat Equation in 1d

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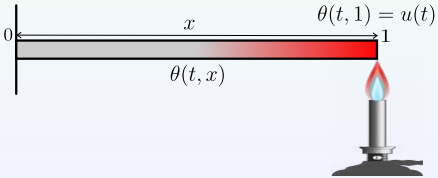


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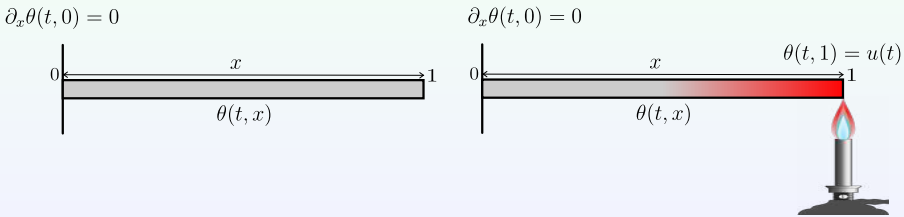
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The Heat Equation in 1d



The heat equation in the interval $(0,1)$ is

$$\begin{cases} \partial_t\theta(t,x) = \partial_{xx}\theta(t,x), & x \in (0,1), \\ \partial_x\theta(t,0) = 0, \quad \theta(t,1) = u(t), & t \in (0,T), \\ \theta(0,x) = \theta_0(x), & x \in (0,1). \end{cases} \tag{1}$$

- θ is the temperature along the rod
- θ_0 is the initial temperature.
- u is the control input.

Let H, Q be Hilbert spaces. Consider an abstract differential system

$$Z'(t) = F(Z(t), u(t)), \ Z(0) = Z_0. \tag{2}$$

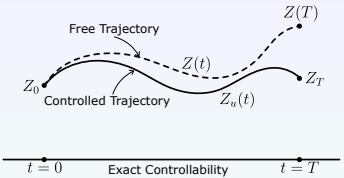
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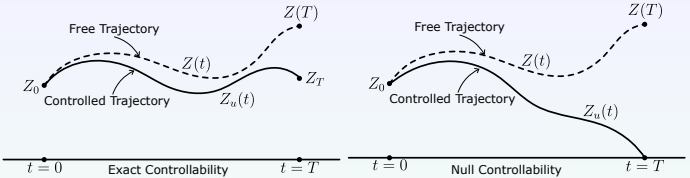


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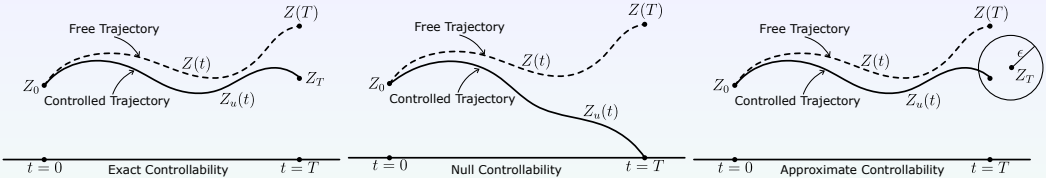


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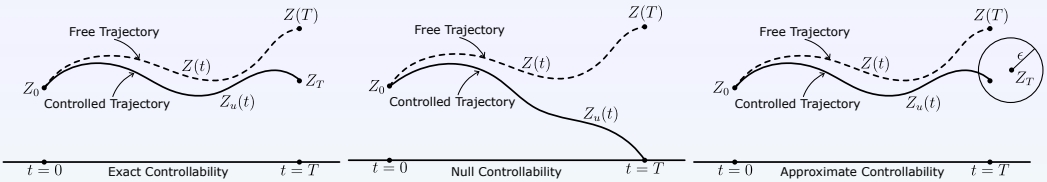


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Aim of Today's Talk

The heat equation (1) is approximately controllable at any time $T > 0$ in the space $L^2(0, 1)$.

The Flatness Method

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$$z'(t) = f(z(t), u(t)), \quad z(t) \in \mathbb{R}^n, \quad u(t) \in \mathbb{R}^m \tag{3}$$

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- We say the system (3) is flat if

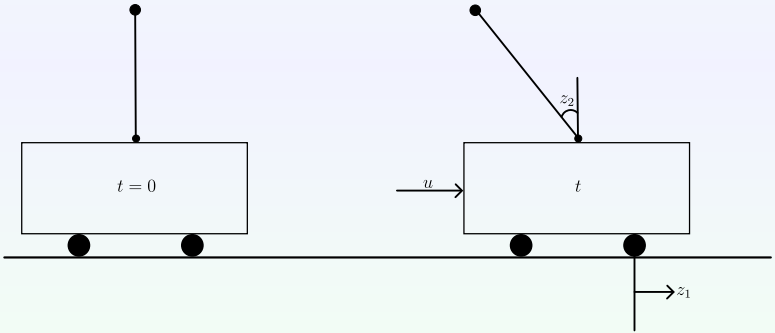
$$z = g\left(y, y', \cdots, y^{(p)}\right), \tag{4}$$

$$u = \varphi\left(y, y', \cdots, y^{(q)}\right), \tag{5}$$

where $p, q \in \mathbb{N} \cup \{0\}$ and g, φ are some smooth functions.

An Illustrative Example

- Consider the linearized inverted pendulum

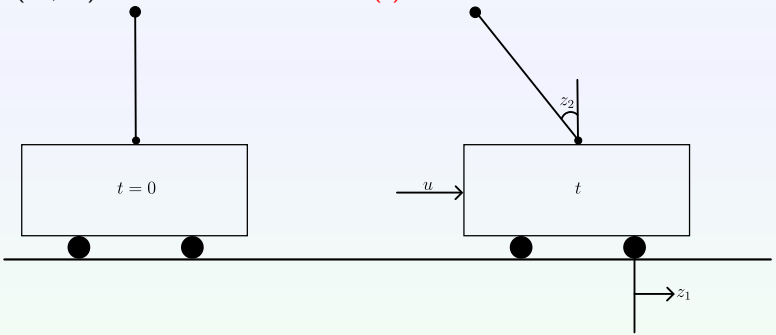


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$$\begin{cases} z_1'(t) = z_2(t), \\ z_2'(t) = u(t), \end{cases} \tag{6}$$

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- One suitable choice is

$$y(t) = \frac{t^2(2T - t)^2}{T^4}, \quad t \in [0, T]. \tag{8}$$

Gevrey Class of Functions

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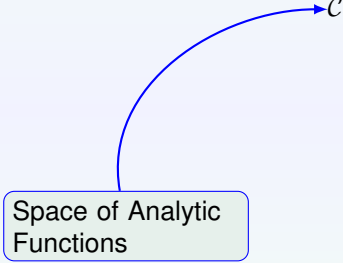
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- Let $\Omega \subset \mathbb{R}$ be an interval and $\alpha > 0$. Then,

$$\mathcal{C}^{\omega}(\Omega) \subset \mathcal{G}^{\alpha}(\Omega) \subset \mathcal{C}^{\infty}(\Omega).$$

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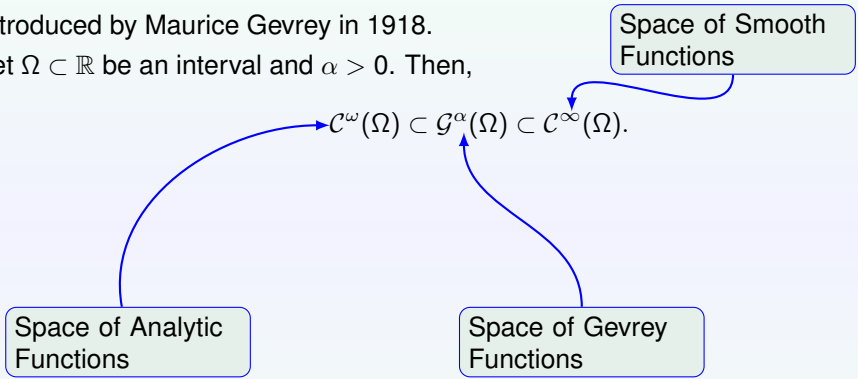
Space of Smooth Functions

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Definition (Gevrey Functions of Order α)

Let $\alpha > 0$. A function $y \in \mathcal{C}^{\infty}[0, T]$ is said to be Gevrey of order α, denoted $y \in \mathcal{G}^{\alpha}[0, T]$, if

$$\exists M, R > 0, \text{ such that } \forall m \in \mathbb{N} \cup \{0\}, \quad \sup_{t \in [0,T]} \left| y^{(m)}(t) \right| \leq M \frac{(m!)^{\alpha}}{R^{m}}.$$

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- Gevrey functions of order 1 are analytic and entire if $0 < \alpha < 1$.
- Gevrey functions of order $\alpha > 1$ have a divergent Taylor expansion.



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Examples

Let $\gamma > 0$. The functions

$$\varphi_\gamma(t) := \begin{cases} \exp\left(\frac{-1}{((1-t)t)^\gamma}\right), & \text{if } t \in (0,1), \\ 0, & \text{if } t \notin (0,1), \end{cases} \quad \text{and} \quad \Phi_\gamma(t) := \begin{cases} \frac{\int_0^t \varphi_\gamma(\tau)d\tau}{\int_0^1 \varphi_\gamma(\tau)d\tau}, & \text{if } t \in (0,1), \\ 0, & \text{if } t \leq 0, \\ 1, & \text{if } t \geq 1, \end{cases}$$

are Gevrey of order $1 + \frac{1}{\gamma}$.

Definition

Let $\alpha, \beta > 0$. A function $\theta \in \mathcal{C}^\infty([0,T] \times [0,1])$ is said to be Gevrey of order α in t and β in x, denoted $\theta \in \mathcal{G}^{\alpha,\beta}([0,T] \times [0,1])$, if $\exists M, R, S > 0$, such that $\forall m, n \in \mathbb{N} \cup \{0\}$

$$\sup_{(t,x)\in[0,T]\times[0,1]} \left| \frac{\partial^{m+n}\theta}{\partial t^m \partial x^n}(t,x) \right| \leq M \frac{(m!)^\alpha (n!)^\beta}{R^m S^n}.$$

Motion Planning of Heat Equation

Formal Solution

- Recall the heat equation

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- The formal control is

$$u(t) = \sum_{k\geq 0} \frac{y^{(k)}(t)}{(2k)!}. \tag{11}$$

Theorem

- If $y \in \mathcal{G}^{\alpha}[0, T]$ with $0 < \alpha < 2$, the formal solution (10) is Gevrey of order α in t and 1 in x.
- In particular, the formal control (11) is Gevrey of order α.

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Proof

Note that

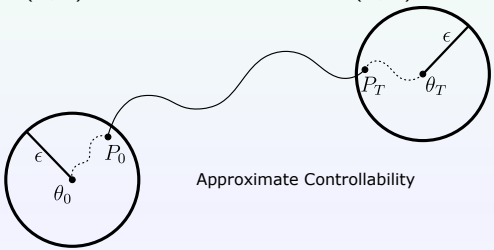
$$\left| \frac{1}{m!^{\alpha} n!} \frac{\partial^{m+n} \theta}{\partial t^m \partial x^n}(t, x) \right| \leq \frac{M}{\tilde{R}^m} \sum_{2k \geq n} \frac{k!^{\alpha-2} \sqrt{k\pi}}{\tilde{R}^k}.$$

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- For every $\epsilon > 0$, there exists polynomials

$$P_0(x) := \sum_{k=0}^{n} p_k^0 \frac{x^{2k}}{(2k)!}, \quad P_T(x) := \sum_{k=0}^{n} p_k^T \frac{x^{2k}}{(2k)!}$$

such that $\|\theta_0 - P_0\|_{L^2(0,1)} < \epsilon$ and $\|\theta_T - P_T\|_{L^2(0,1)} < \epsilon$.

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- The function

$$Y(t) := \sum_{k=0}^{n} p_k^0 \frac{t^k}{k!} \left(1 - \Phi_\gamma\left(\frac{t}{T}\right)\right) + p_k^T \frac{(T-t)^k}{k!} \Phi_\gamma\left(\frac{t}{T}\right)$$

is Gevrey of order $0 < \alpha < 2$ when $\gamma > 1$ and satisfies

$$\begin{cases} Y^{(k)}(0) = p_k^0, \quad Y^{(k)}(T) = p_k^T, \quad k = 0, 1, \ldots, n \\ Y^{(k)}(0) = 0, \quad Y^{(k)}(T) = 0, \ \forall k > n. \end{cases}$$

Approximate Controllability

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Theorem

- The control U exactly steers system (1) from the initial state P_0 at time $t = 0$ to the final state P_T at time $t = T$.
- The approximate control $\bar{U}$ approximately steers system (1) from the initial state θ_0 to the final state θ_T.

Usefulness of Flatness Method

- Controllability of ODEs, Heat Equation, Korteweg-de Vries (KdV) equation, Schrödinger equation and some coupled ODE-Parabolic systems.

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Thank You for Your Attention ...

